SOURAV KUMAR

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PROFESSIONAL SUMMARY

AI Engineer (fresher) with proven experience designing and shipping production-grade intelligent systems. Built Enterprise-FraudShield-AI — a real-time fraud detection backend processing 10,000+ transactions/sec using FastAPI, Kafka, Redis, and XGBoost — and developed LLM-powered applications leveraging LangChain, RAG pipelines, and Hugging Face Transformers. Strong foundation in Python, deep learning, NLP, and cloud deployment (AWS, Docker). IBM-certified in Machine Learning; additional credentials from Harvard CS50AI and Microsoft Azure AI-900. Ready to contribute to AI product teams building scalable, real-world AI solutions from day one.

TECHNICAL SKILLS

Languages	Python, Java, Kotlin, JavaScript, HTML/CSS, SQL, C, C++
AI / LLM / NLP	LangChain, RAG, LLM, Hugging Face, Transformers, Fine-tuning, Prompt Engineering, NLP, OpenCV, YOLO, ResNet, MobileNetV3
ML Frameworks	Scikit-learn, XGBoost, TensorFlow, LSTM, ARIMA, Prophet, Linear Regression, Decision Tree, Random Forest
Data and Analysis	Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Hyperparameter Tuning, Cross-validation
Backend and APIs	FastAPI, Flask, Django, Node.js, REST APIs, Microservices, Async Processing
Cloud and DevOps	AWS (S3, EC2, Lambda), Docker, Jenkins, Apache Kafka, Redis, Git, GitHub
Model Evaluation	MAE, MSE, RMSE, R2 Score, Confusion Matrix, ROC-AUC

EXPERIENCE

Machine Learning Intern

IBM | Remote | Jun 2025 – Jul 2025

- Preprocessed and cleaned 10,000+ structured healthcare records, reducing data inconsistencies by 30% and improving downstream model accuracy.
- Performed exploratory data analysis (EDA) to identify top cost-driving factors including BMI, age, and smoking status across patient segments.
- Developed and compared regression models (Linear Regression, Decision Tree, Random Forest), achieving 87% prediction accuracy on medical insurance charges.
- Built end-to-end predictive pipeline from raw data ingestion to model deployment using Python, Pandas, and Scikit-learn.
- Visualised key insights and model performance using Matplotlib and Seaborn, presenting findings to the project team.

Social Work Intern

Samadhaan Foundation | On-site | Jun 2024 – Aug 2024

- Assisted in community outreach programs serving 200+ beneficiaries.
- Conducted data analysis and supported digital management initiatives to improve organisational efficiency.

PROJECTS

Enterprise-FraudShield-AI

Python, FastAPI, Kafka, Redis, XGBoost, Docker | Dec 2024

- Engineered a real-time fraud detection system processing 10,000+ transactions/sec, reducing detection latency by 60% using FastAPI, Apache Kafka, and Redis.

- Implemented ML models (Scikit-learn, XGBoost, anomaly detection) achieving 94% fraud detection accuracy on financial transaction datasets.
- Built production-ready microservices backend with Docker and Uvicorn, enabling scalable deployment with asynchronous REST API endpoints.
- Integrated behavioural analytics pipeline to identify refund abuse and suspicious patterns across distributed enterprise data streams.

Stock Price Prediction Models

Python, LSTM, ARIMA, Prophet, Matplotlib | Aug 2025

- Developed and benchmarked LSTM, ARIMA, and Prophet models for stock price forecasting, improving prediction accuracy by 22% through ensemble comparison.
- Performed hyperparameter tuning and cross-validation to reduce overfitting, achieving stable forecasting outcomes across 5 different stock datasets.
- Built interactive EDA dashboards using Matplotlib and Seaborn to visualise historical trends, volatility, and model performance metrics.

Fish Species Classifier (Transfer Learning)

Python, MobileNetV3, TensorFlow, OpenCV | May 2024

- Trained image classification model using MobileNetV3-Large on 9,000 images across 9 fish species, achieving 90% overall accuracy.
- Built data augmentation and preprocessing pipelines (80/20 train-validation split) ensuring robust generalisation across varied datasets.
- Eliminated manual inspection bottleneck, reducing classification time by 85% compared to manual review.

EDUCATION

B.Tech, Computer Science and Engineering (AI/ML)

University of Petroleum and Energy Studies (UPES), Dehradun | Expected May 2026

CERTIFICATIONS

- Harvard CS50AI: Introduction to Artificial Intelligence with Python — 2024
- Microsoft Certified: Azure AI Fundamentals (AI-900) — 2024
- IBM Certified: Healthcare Cost Prediction using Machine Learning — 2025
- Data Structures and Algorithms — Coursera — 2025
- Python Programming — CodeWithHarry — 2023

ACHIEVEMENTS AND ACTIVITIES

- Led team strategy and rapid prototyping at Smart India Hackathon 2024, building scalable real-world AI solutions.
- Won prize at CSI Hackathon (2x participant) for delivering innovative team-based AI product solutions.
- Active competitive programmer on LeetCode and CodeChef — consistently solving algorithmic problems under timed conditions.
- Member, Coding Club UPES (2022–2026) — contributed to organising events and representing the club in competitions.
- Volunteered at TechFest 2023, UPES — assisted in event coordination and management.